USER INSTRUCTIONS
BUS PLAN



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Introduction

In this user guide, we will introduce you to the tool that we developed for Transdev. The tool checks whether a bus schedule is correct and feasible. In this user manual, we will mainly focus on bus lines 400 and 401.

The goal of the tool is to improve the bus schedules by reducing costs and the number of employees required. We will explain step by step how the tool works, which data it uses, how to interpret the improved bus schedule and how to understand the other information provided by the tool.

The tool can be accessed via the following URL:

https://transdev-busplan-tool.streamlit.app/

Instructions

0. Starting the app

To start the app, you have to click on the link in the *introduction* section. It could happen that you will receive the next message in figure 1. If you click on “Yes, get this app back up!”, the app will load and start automatically.

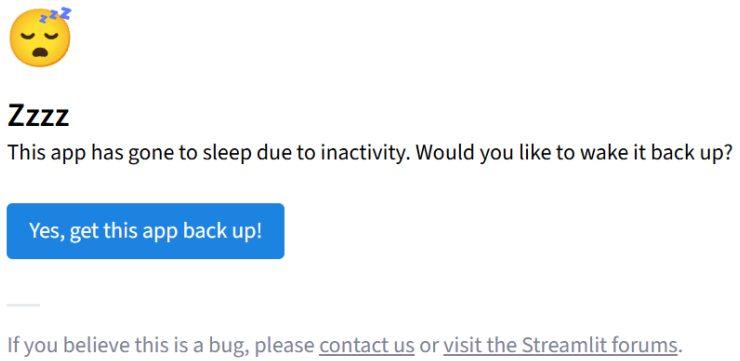


Figure 1 Message when clicking on the URL

If you didn’t receive the message in figure 1. You should see the starting screen of the app. In figure 2 is shown, what you should see from the app.

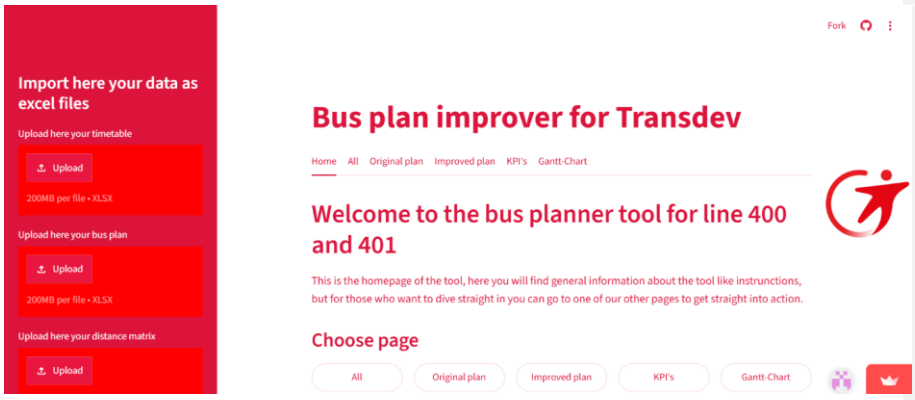


Figure 2 Startscreen of the app

1. Import the data

When you have reached the homepage of the app. You will see on the left side of your screen three light red boxes who are asking for a file. You need to deliver three excel files into the app: timetable, bus plan and a distance matrix.

- The timetable is the planning of the bus line(s).
- The bus plan is the excel document where you can see where the start- and end locations are, the kind of trip the bus makes, the bus number and the energy consumption of the bus.
- The distance matrix is the excel document where the distances and times are in from the start- and end locations.

When you have the three specified excel files ready, you can start uploading them your searches will open atomically. You can drag or select the specific file to each upload field. Note that you have to upload the three excel files separately. In figure 3 is an example how to upload the excel files. See the appendix for the data formats for each file.

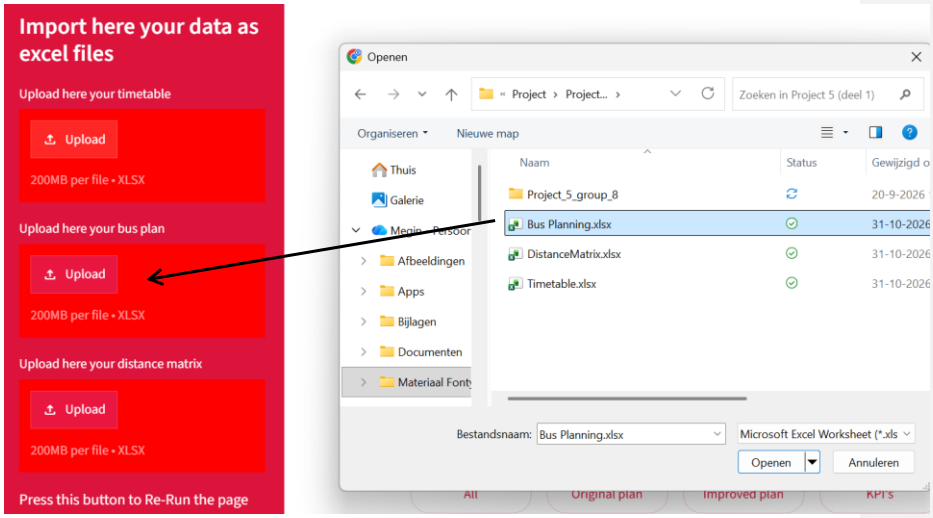


Figure 3 You can drag or select your Excel file to the right upload field

Note that every time you are uploading and/or clicking on something, a person is taking some poses, see figure 4. This means that the app is loading. Do not switch tabs or pages when loading. This will slow the page.



Figure 4 Person taking a pose

Commented [FT1]: "When you got all three excel files, you can click on upload, your searches will open atomically. " je kan dit misschien iets anders verwoorden zoals "when you have the three specified excel files ready, you can start uploading them"

Commented [FT2]: Je kan hierbij op het einde ook zeggen dat je die bar tijdelijk kan verstoppen om de rest van de pagina beter zichtbaar te maken

Commented [JD3]: Ik heb nog even een belangrijk zinnetje onderaan toegevoegt...

Commented [FT4]: "red boxes who are asking" kan je veranderen naar "that are asking' denk ik.

2. Pages: the *original plan* and the *improved plan* (questions)

When you have uploaded the three files and the app is done loading. You can go through the pages, see figure 5. You start automatically at the home page. We will skip the page "all", because on this page everything is shown in order. We will start with the page "original plan" and "improved plan", because they show the same information. The only difference between the two pages is, that the original bus plan your bus plan is that you uploaded and the improved bus plan is the improved version of your original bus plan



Figure 5 Pages of the tool

Both pages start with two questions: "Is the bus plan feasible?" and "Does the bus plan schedule all trips from the timetable?"

In the example we use, the original bus plan is not feasible. You can see in which rows the errors are in. When you click on the rows, for example *[0 – 99]*, there will be some explaining why the bus plan is not feasible. In our case we clicked on the next arrow, and we see that there is overlap in the times of bus 7, as you can see in figure 6.



Figure 6 Example of the first question when it is false

If there are no errors, the answer of the question will be *True*.

Commented [FT5]: “are riding” naar “are driving” verandert

3.

Pages: the *original plan* and the *improved plan* (visual)

When you scroll more down on the page, you will see a Gantt-chart, this is a visualisation of the bus plan. You can see how many busses are driving, which time they are driving, which trip they make and when they are idle. The matching colours with the trips and idling are in the legend beneath the Gantt-chart. The whole picture could be a bit overwhelming. Above the Gantt-chart you see the numbers of the busses. When you don’t want to see some busses, you click on the cross next to the number. In the figure 7, we crossed of all the lines in our example bus plan, until we only see bus 1 to 5.

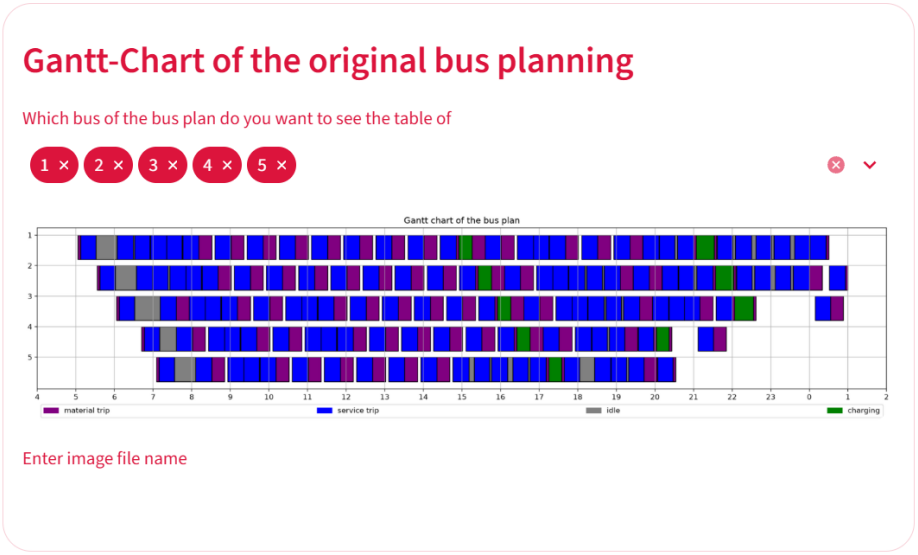


Figure 7 Example when you want to see less buslines in a visualisation

An extra option when you reached the schedule of the lines that you want to see, you can type under “Enter image file name” a file name of the picture to download it.

When you want to compare these two visuals on the same page, you can go to the page: *Gantt-chart*. You can do the same things as described earlier, but it is easier to compare one line in both visualisations.

4. Pages: the *original plan* and the *improved plan* (KPI's)

On the end of the pages, you can see a table with some important information about the (improved) bus plan, depending on which page you are looking. There will be three aspe shown in the table:

1. Total busses: the number of busses that are needed for the bus plan.
2. Total energy consumption: how much energy (kWh) all the busses have used together in one day of the bus plan.
3. Idling time (time not moving): the time that the busses are standing still.

In figure 8 is shown how the table looks in our example.

	Original plan
Total busses	20
Total energy consumption	6378.060399999999
Time not moving	0 days 21:20:00

Figure 8 Table with three important aspects for the bus plan

When you click on the page "KPI's". The three aspects in the table of the original and improved plan are showed, see figure 9. You can easily compare your original bus plan with your improved bus plan as you can see in our example. We can conclude in this example that our improved bus plan uses 3 busses less then in our original plan, but the idle time is way more.

	Original plan	Improved plan
Total busses	20	17
Total energy consumption	6378.060399999999	6170.9197
Time not moving	0 days 21:20:00	2 days 19:44:00

Figure 9 Comparing KPI's

Commented [FT6]: Wat wil je hier mee zeggen "There will be three aspect shown in the table:", want dat is niet helemaal duidelijk voor mij

Commented [FT7]: "amount" naar "number" verandert

Commented [FT8]: "all the have used in one day of the bus plan.". Wat probeer je daarmee te zeggen, ik denk dat je iets met bussen daarmee wil zeggen

5.

Downloading the improved bus plan

Commented [FT9]: Hierbij kan je misschien een plaatje geven waarbij je kan zien waar op de pagina het downloaden nou zit

So far we have discussed how to see if your bus plan is feasible, a visualisation of the bus plan and some important KPI's. We also showed how to compare the visualisations and the KPI"s with the original and improved bus plan.

We got an option to download your improved bus plan in an Excel file, in the same format as you gave your original plan. To download your improved bus plan. You should scroll down on the left side on the page in the red side, until you see the same text as in figure 10. To download your improved bus plan, you have to give the bus plan a name, click on enter and you should be able to download it.

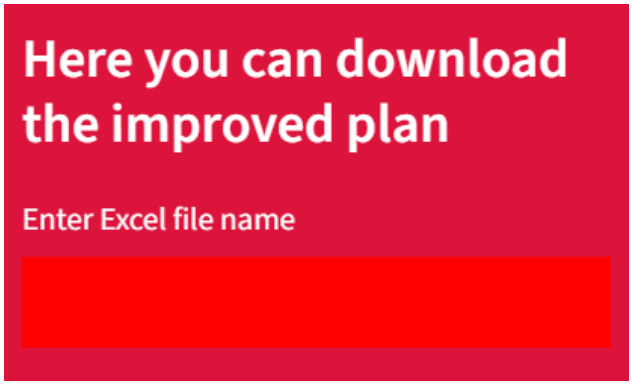


Figure 10 Field to enter a Excel file name for your improved bus plan

Appendix

The data in the Excel files must have a specific data type. This means that it is important how the data is noted in the Excel file. In the next three figures is shown what the data has to be for all the columns in the Excel files that has to be imported in the tool for an improved bus plan.

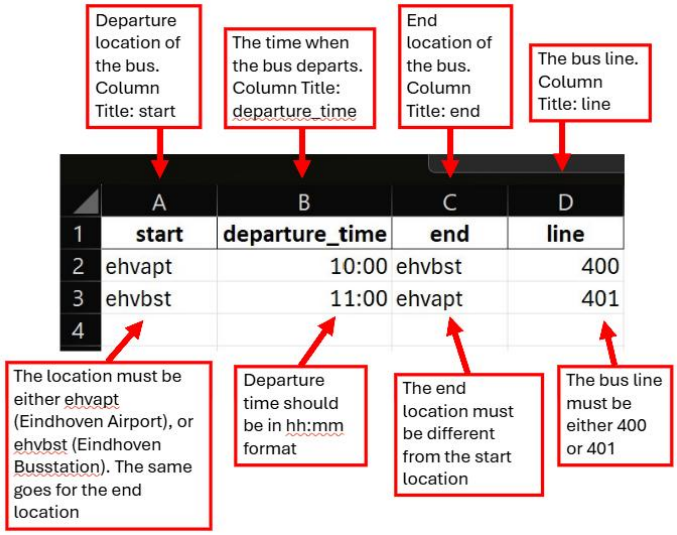


Figure 11 Description of the data of the timetable Excel file

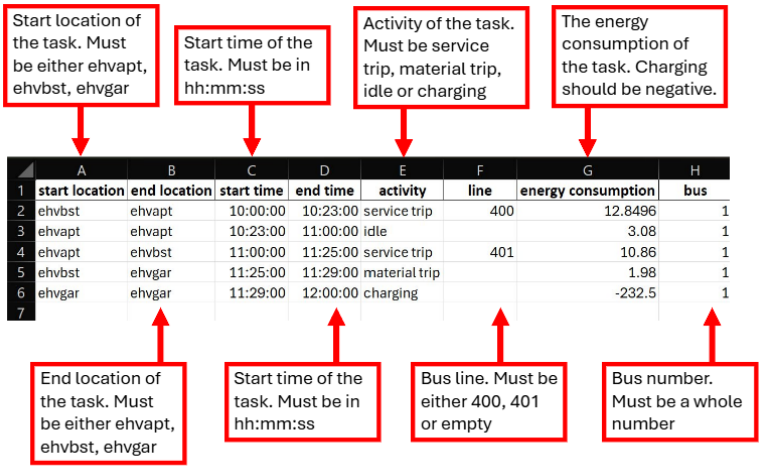
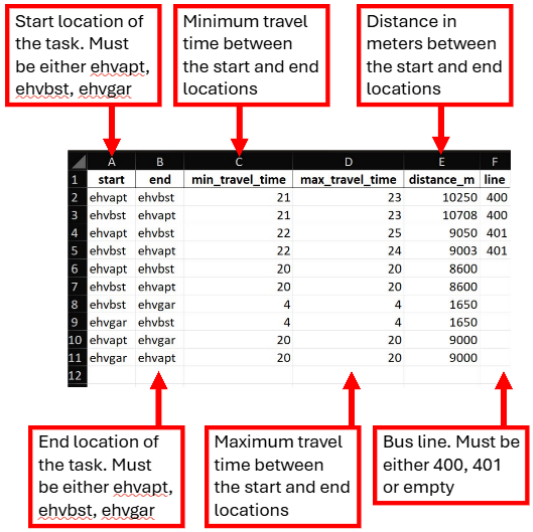


Figure 12 Description of the data of the bus plan Excel file



	A	B	C	D	E	F
1	start	end	min_travel_time	max_travel_time	distance_m	line
2	ehvapt	ehvbst	21	23	10250	400
3	ehvbst	ehvapt	21	23	10708	400
4	ehvapt	ehvbst	22	25	9050	401
5	ehvbst	ehvapt	22	24	9003	401
6	ehvapt	ehvbst	20	20	8600	
7	ehvbst	ehvapt	20	20	8600	
8	ehvbst	ehvgar	4	4	1650	
9	ehvgar	ehvbst	4	4	1650	
10	ehvapt	ehvgar	20	20	9000	
11	ehvgar	ehvapt	20	20	9000	
12						

Figure 13 Description of the data of the distance matrix Excel file